Cubic Graphs

Exam code: 0607

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Cubic Graphs

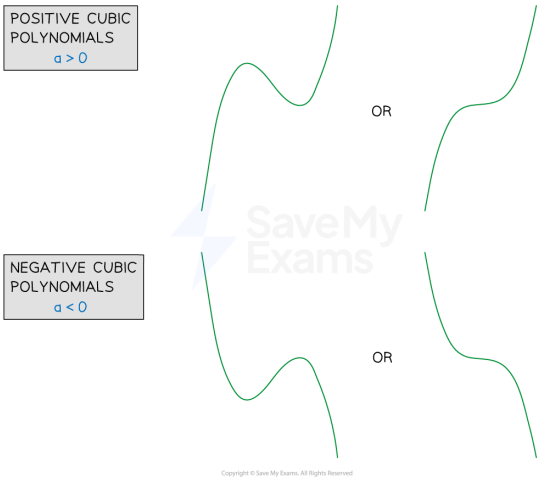
What is a cubic?

- A **cubic** is a function of the form $ax^3 + bx^2 + cx + d$
  - a, b, c and d are constants
  - It is a polynomial of **degree** (order) 3
    - So b, c and/or d could be zero
    - but a cannot be equal to zero
- To sketch the graph of a cubic, the polynomial needs to be in **factorised form**
  - E.g. $(2x-1)(x+2)(x-3)$ is the factorised form of $2x^3 - 3x^2 - 11x + 6$
- You do **not** need to **factorise a cubic** unless it is a more simple one
  - E.g. $x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2)$
  - You should be able to **expand three brackets** to find the expanded form of a cubic

Help me understand

What does the graph of a cubic look like?

- In general the graph of a cubic will take one of the four forms
  - All are **smooth** curves

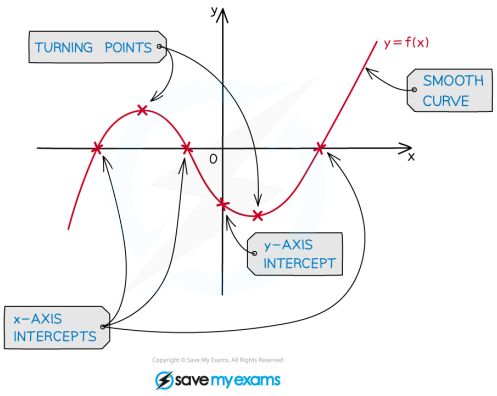


Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The exact form of a particular cubic will depend on:

        - If $a > 0$ the graph is a **positive cubic** ('starts' in the bottom left, 'ends' in the top right)
        - If $a < 0$ the graph is a **negative cubic** ('starts' in the top left, 'ends' in the bottom right)

    - The **turning points**



- Cubics can have **two turning points**
    - a maximum point and a minimum point

- However, note that the graphs of $y = x^3$ and $y = -x^3$:
    - Do not have a maximum or minimum (turning points)
    - Only cross the x-axis once, at $x = 0$

How do I sketch the graph of a cubic?

- **STEP 1**
  Find the y**-axis intercept** by setting $x = 0$
- **STEP 2**
  Find the x-axis intercepts (**roots**) by setting $y = 0$
    - In factorised form, this can be done by **inspection**
        - A cubic of the form $y = (x - p)(x - q)(x - r)$ has roots at p, q and r
        - E.g. $y = (x - 2)(x - 3)(x + 5)$ has roots at 2, 3, and -5

    - Any **repeated roots** will mean the graph **touches** the x-axis
        - The graph does **not cross** the x-axis
        - E.g. $(x - 2)^2(x + 1)$ **touches** the x-axis at $x = 2$, and **intersects** the x-axis at $x = -1$

- **STEP 3**
  Consider the **shape** of the graph
    - Is it a **positive cubic** or a **negative cubic**?
    - Where does the graph 'start' and 'end'?

- **STEP 4**
  Consider where any **turning points** should go
- **STEP 5**
  Sketch the graph with a **smooth curve**
  Label points where the graph **intercepts** the x and y axes

Sketch the graph of $y = (2x - 1)(x - 3)^2$.

STEP 1
Find the y-axis intercept by substituting in $x = 0$

$$y = (-1)(-3)^2 = -9$$

STEP 2
Find the x-axis intercepts by solving $y = 0$
Either bracket can be equal to zero

$$(2x - 1) = 0$$
$$x = \frac{1}{2}$$

$$(x - 3)^2 = 0$$
$$x = 3$$

(repeated solution, as there are two $(x - 3)$ brackets

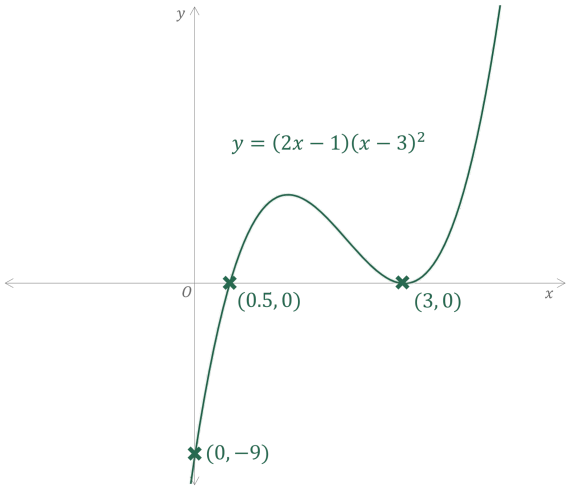
STEP 3
Consider the shape, and the 'start' and 'end' points:

$a > 0$ $(a = 2)$ so it is a positive cubic
$x = 3$ is a repeated root so the graph will *touch* the x-axis at this point

STEP 4
Consider the turning points

One turning point (minimum) will need to be where the curve touches the x-axis
The other (maximum) will need to be between the two roots $x = \frac{1}{2}$ and $x = 3$

STEP 5
Sketch a smooth curve with labelled intercepts



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